



AUO Display+

Product Specification

G150XTN06.8

☐ Preliminary Specification

☒ Final Specification

Module	15 Inch Color TFT-LCD
Model Name	G150XTN06.8

Company	
Checked & Approved by	Date

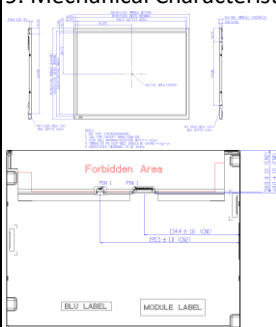
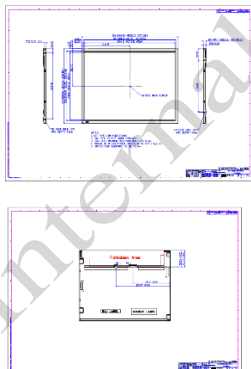
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General Display Business Unit / AUO Display Plus Corporation	

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Record of Revision

Version and Date	Page	Old description	New Description																																																																						
0.0 2016/11/10	All	First Edition																																																																							
1.0 20170523	5	2.1 Display Characteristics Typical Power Consumption : Support color : 16.2M/262	2.1 Display Characteristics Typical Power Consumption : 9.78W Support color : 6+2FRC(16.2M)																																																																						
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	Off Control Voltage ²⁾	— ³⁾	— ⁴⁾	0.7 ⁵⁾	Volt ⁶⁾																																																																																																																																																																																						
F _{PWM} ¹⁾	PWM Dimming Frequency ²⁾	200 ³⁾	— ⁴⁾	20k ⁵⁾	Hz ⁶⁾																																																																																																																																																																																						
	High Voltage ²⁾	2.5 ³⁾	3.3 ⁴⁾	5.5 ⁵⁾	Volt ⁶⁾																																																																																																																																																																																						
	Low Voltage ²⁾	— ³⁾	— ⁴⁾	0.7 ⁵⁾	Volt ⁶⁾																																																																																																																																																																																						
	Dimming Duty Cycle ²⁾	5 ³⁾	— ⁴⁾	100 ⁵⁾	% ⁶⁾																																																																																																																																																																																						
I _F ¹⁾	LED Forward Current ²⁾	— ³⁾	64 ⁴⁾	— ⁵⁾	mA ⁶⁾	Ta = 25°C ⁷⁾																																																																																																																																																																																					
LED Life ¹⁾	— ²⁾	50,000 ³⁾	— ⁴⁾	— ⁵⁾	Hrs ⁶⁾	Ta = 25°C ⁷⁾																																																																																																																																																																																					
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V _{CC} ¹⁾	Input Voltage ²⁾	10.8 ³⁾	12 ⁴⁾	13.2 ⁵⁾	Volt ⁶⁾																																																																																																																																																																																						
I _{VCC} ¹⁾	Input Current ²⁾	— ³⁾	0.58 ⁴⁾	0.65 ⁵⁾	A ⁶⁾	100% Dimming ⁷⁾																																																																																																																																																																																					
P _{LED} ¹⁾	Power Consumption ²⁾	— ³⁾	6.98 ⁴⁾	7.80 ⁵⁾	Watt ⁶⁾	100% Dimming ⁷⁾																																																																																																																																																																																					
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24	9. Mechanical Characteristics						9. Mechanical Characteristics																																																																																																																																																																																				
																																																																																																																																																																																											

1. Operating Precautions

- 1) Display area (Polarizer) of TFT-LCD Module is easily to be damaged, please be cautious and not to scratch it.
- 2) Be sure to power off your machine before connecting or disconnecting your signal cable to TFT-LCD Module.
- 3) Wipe off water drop on display area immediately. Long contact with water may cause discoloration or spots.
- 4) When the panel surface is soiled, wipe it with absorbent cotton or soft cloth.
- 5) Display area (Glass) of TFT-LCD Module may be broken or cracked if bump Module against hard object.
- 6) To avoid ESD (Electro Static Discharge) damage, be sure to ground yourself before handling TFT-LCD Module.
- 7) Do not open nor modify the TFT-LCD module assembly.
- 8) Do not press the reflector sheet at the back of the module to any direction.
- 9) In case if TFT-LCD module has to be put back into the packing container slot after it was taken out from the container, do not press the center of the LED Reflector edge. Instead, press at the far ends of the LED Reflector edge softly. Otherwise the TFT-LCD Module may be damaged.
- 10) When inserting or removing of your signal cable to TFT-LCD Module, be sure not to apply abnormal force (rotate, tilt...etc.) to the Connector of the TFT-LCD Module.
- 11) TFT-LCD Module is not allowed to be twisted & bent even force is added on module in a very short time. Please design your display product well to avoid external force applying to module by end-user directly.
- 12) Small amount of materials without flammability grade are used in the TFT-LCD module. The TFT-LCD module should be supplied by power complied with requirements of Limited Power Source (IEC60950 or UL1950), or be applied exemption.
- 13) Severe temperature condition may result in different luminance, response time.
- 14) Continuous operating TFT-LCD Module under high temperature environment may accelerate LED light bar exhaustion and reduce luminance dramatically.
- 15) The data on this specification sheet is applicable when TFT-LCD module is placed in landscape position.
- 16) Continuous displaying fixed pattern may induce image sticking. It's recommended to use screen saver or moving content periodically if fixed pattern is displayed on the screen.

2. General Description

G150XTN06.8 is a Color Active Matrix Liquid Crystal Display composed of a TFT-LCD display, a driver circuit, and a backlight system. The screen format is intended to support XGA (1024(H) x 768(V)) screen and 16.2M or 262K colors. All input signals are LVDS interface compatible. All design rules of this module can correspond to PSWG standard.

G150XTN06.8 is designed for industrial display applications.

2.1 Display Characteristics

The following items are characteristics summary under 25°C condition. (Room Temperature)

Items	Unit	Specifications
Screen Diagonal	inch	15
Active Area	mm	304.128(H) x 228.096(V)
Pixels H x V		1024x3(RGB) x 768
Pixel Pitch	mm	0.297 x 0.297
Pixel Arrangement		R.G.B. Vertical Stripe
Display Mode		TN, Normally White
Nominal Input Voltage VDD	Volt	3.3 typ.
Typical Power Consumption	Watt	9.28W
Weight	Grams	1000g (max.)
Physical Size	mm	326.5(H)x 253.5(V) x 10.0(D) (max.)
Electrical Interface		LVDS
Surface Treatment		Anti-glare
Support Color		6+2FRC (16.2M)
Temperature Range		
Operating	°C	-30 to +85
Storage (Non-Operating)	°C	-30 to +85
RoHS Compliance		RoHS Compliance

2.2 Optical Characteristics

The optical characteristics are measured under stable conditions at 25°C (Room Temperature).

Item	Unit	Conditions	Min.	Typ.	Max.	Note
White Luminance	[cd/m2]		360	450	-	1
Uniformity	%	9 Points	75	80	-	1, 2, 3
Contrast Ratio			400	800	-	4
Cross talk	%		-	1.2	1.5	5
Response Time	[msec]	Rising	-	5.7		6
	[msec]	Falling	-	2.3		
	[msec]	Raising + Falling	-	8		
Viewing Angle	[degree]	Horizontal (Right)	70	80	-	7
	[degree]	CR = 10 (Left)	70	80	-	
	[degree]	Vertical (Upper)	60	70	-	
	[degree]	CR = 10 (Lower)	70	80	-	
Color / Chromaticity Coordinates (CIE 1931)		Red x	0.547	0.597	0.647	
		Red y	0.289	0.339	0.389	
		Green x	0.279	0.329	0.379	
		Green y	0.546	0.596	0.646	
		Blue x	0.104	0.154	0.204	
		Blue y	0.003	0.053	0.103	
		White x	0.263	0.313	0.363	
		White y	0.279	0.329	0.379	
Color Gamut	%			60	-	

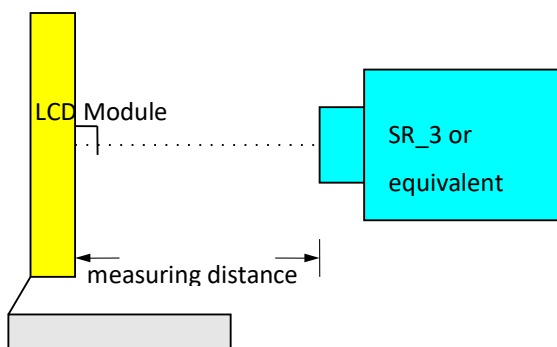
Note 1: Measurement method

Equipment Pattern Generator, Power Supply, Digital Voltmeter, Luminance meter (SR_3 or equivalent)

Aperture 1° with 50cm viewing distance

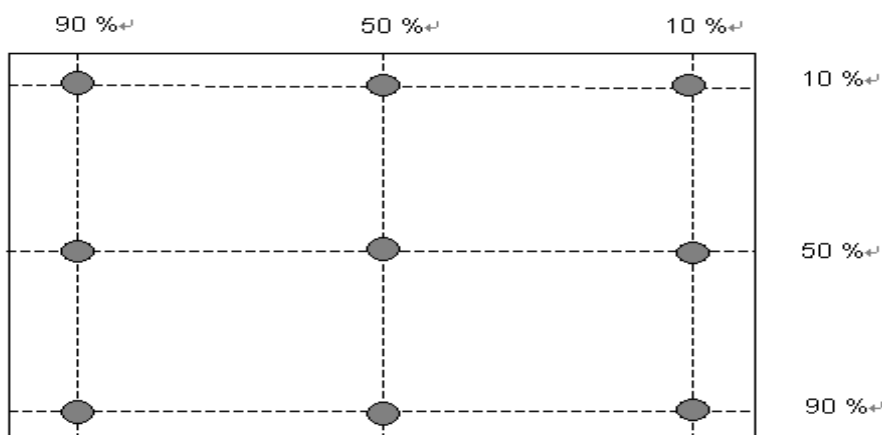
Test Point Center

Environment < 1 lux



Module Driving Equipment
Proprietary

Note 2: Definition of 9 points position (Display active area: 304.128(H) x 228.096(V))



Note 3:

The luminance uniformity of 9 points is defined by dividing the maximum luminance value by the minimum luminance value at full white condition.

$$\delta W5 = \frac{\text{Maximum Brightness of Nine points}}{\text{Minimum Brightness of Nine points}}$$

Note 4: Definition of contrast ratio (CR):

$$\text{Contrast ratio (CR)} = \frac{\text{Brightness @ "White" state}}{\text{Brightness @ "Black" state}}$$

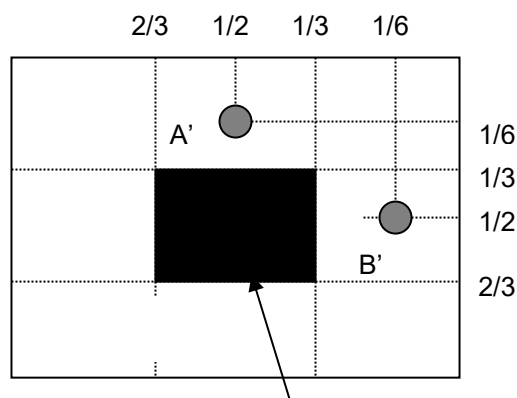
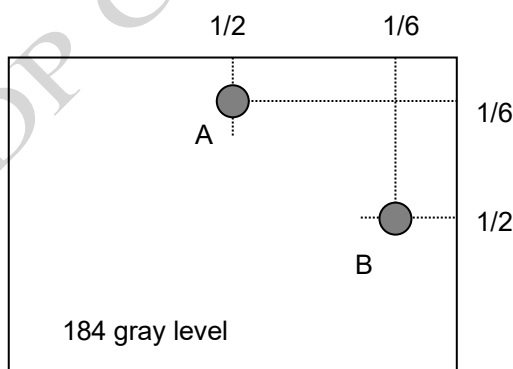
Note 5 : Definition of cross talk (CT)

$$CT = | YB - YA | / YA \times 100 (\%)$$

Where

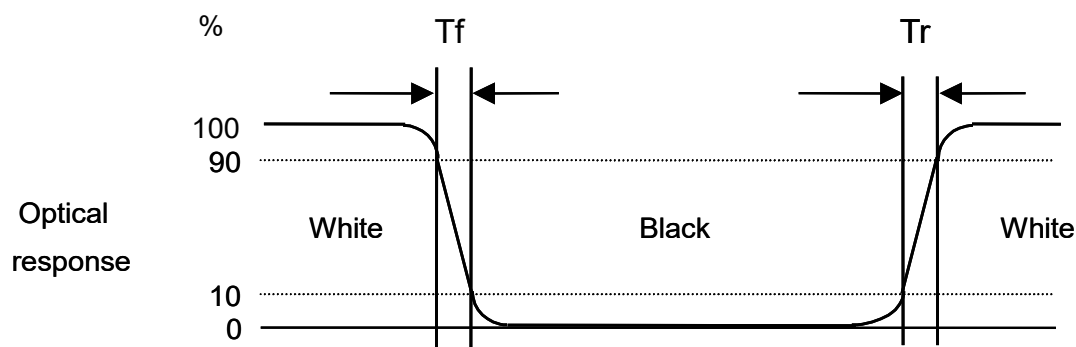
YA = Luminance of measured location without gray level 0 pattern (cd/m2)

YB = Luminance of measured location with gray level 0 pattern (cd/m2)



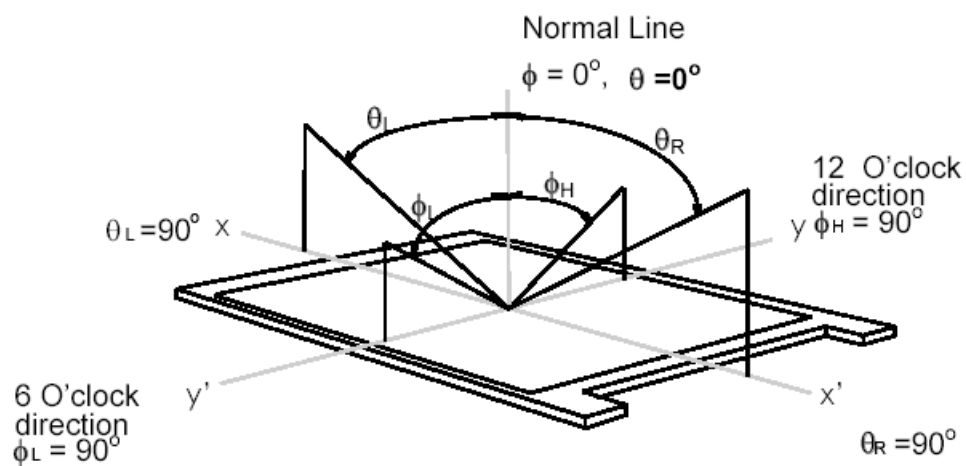
Note 6: Definition of response time:

The output signals of photo detector are measured when the input signals are changed from "White" to "Black" (falling time) and from "Black" to "White" (rising time), respectively. The response time interval definition is between 10% and 90% of amplitude. Please refer to the figure as below.



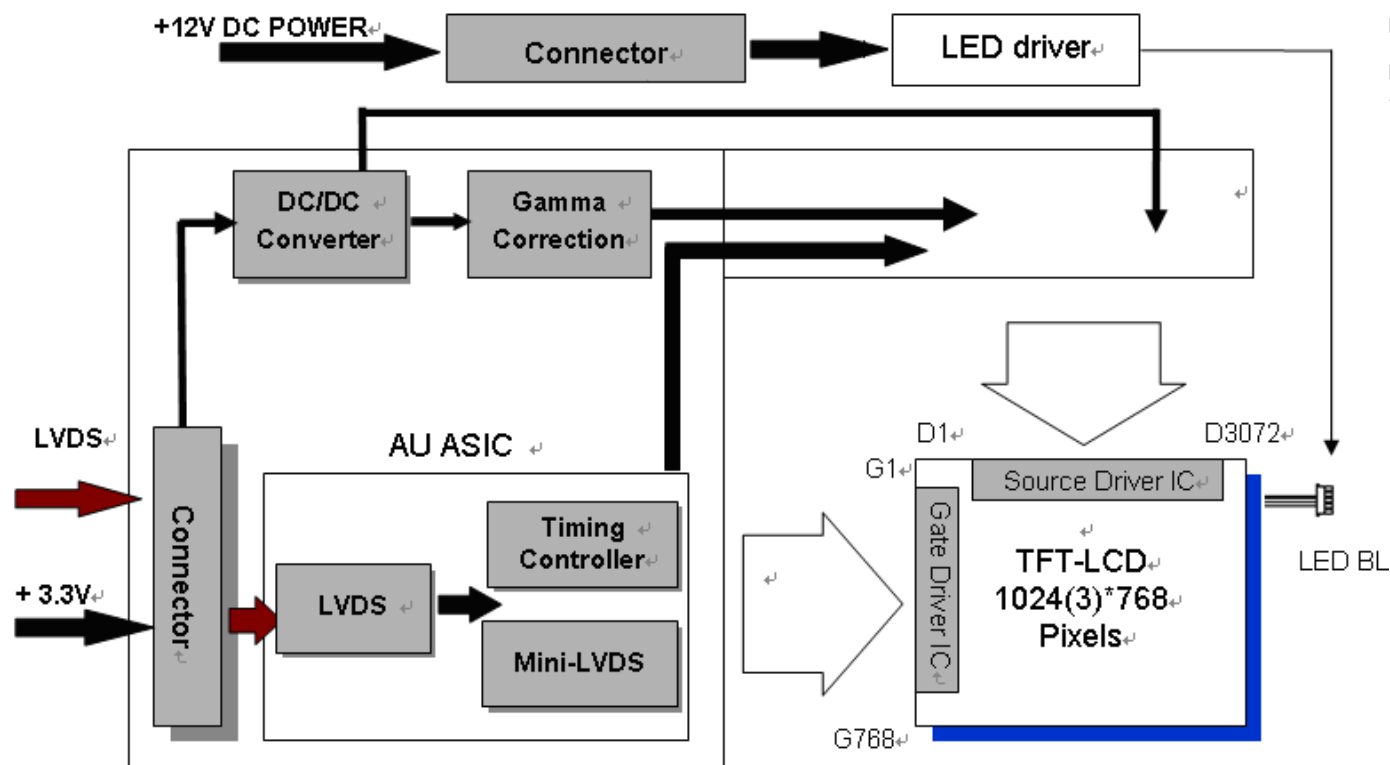
Note 7: Definition of viewing angle

Viewing angle is the measurement of contrast ratio ≥ 10 , at the screen center, over 180° horizontal and 180° vertical range. The 180° horizontal (θ_L , θ_R) and 180° vertical (ϕ_H , ϕ_L) range are illustrated as following figure.



3. Functional Block Diagram

The following diagram shows the functional block of the 15inch color TFT/LCD module.



4. Absolute Maximum Ratings

4.1 Absolute Ratings

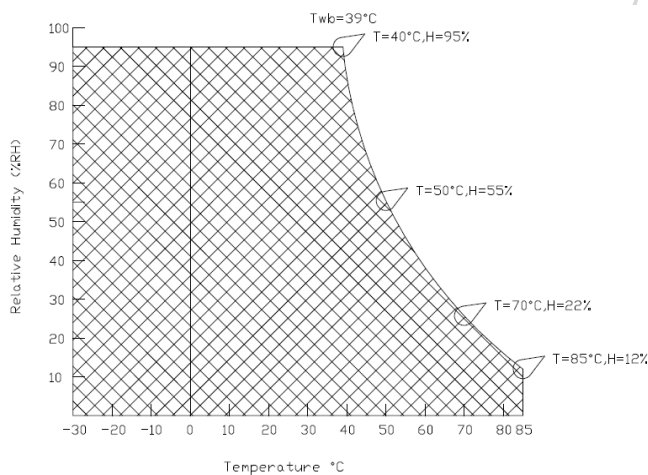
Item	Symbol	Min	Max	Unit	Remark
Logic/LCD Drive Voltage	VDDA	-0.3	3.6	Volt	Ta= 25°C

4.2 Absolute Ratings of Environment

Item	Symbol	Min	Max	Unit	Remark
Operating Temperature	TOP	-30	+85	°C	Note 1, 2
Operation Humidity	HOP	8	90	%RH	Note 1, 2
Storage Temperature	TST	-30	+85	°C	Note 1
Storage Humidity	HST	8	90	%RH	Note 1

Note 1: Permanent damage to the device may occur if exceed maximum values.

Note 2: Maximum Wet-Bulb should be 39°C and no condensation.



Note 3: Only operation is guaranteed. Optical performance should be evaluated at 25°C only.

Note 4: Operation Temperature + 85°C is defined as panel surface temperature.

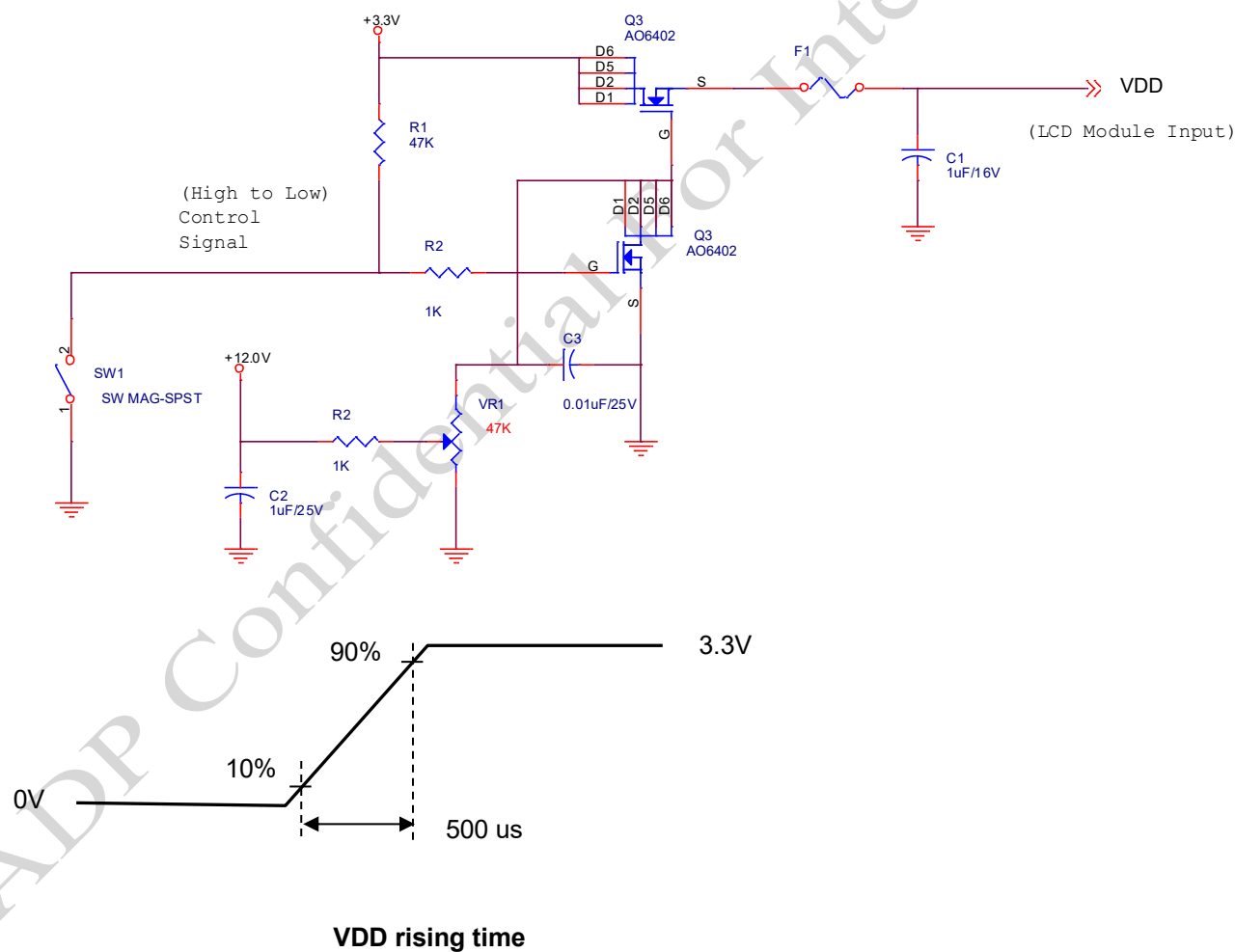
5. Electrical Characteristics

5.1 TFT-LCD Driving

5.1.1 Power Specification

Symbol	Parameter	Min	Typ	Max	Units	Remark
VDD	Logic/LCD Drive Voltage	3.1	3.3	3.6	[Volt]	±10%
IDD	VDD Current	-	0.7	1	[A]	Black pattern. VDD=Typ.3.3V, at 60Hz
Irush	LCD Inrush Current	-	-	3	[A]	Note 1
PDD	VDD Power	-	2.3	3.3	[Watt]	Black pattern. VDD=Typ.3.3V, at 60Hz

Note 1: Measurement condition:



5.1.2 Signal Electrical Characteristics

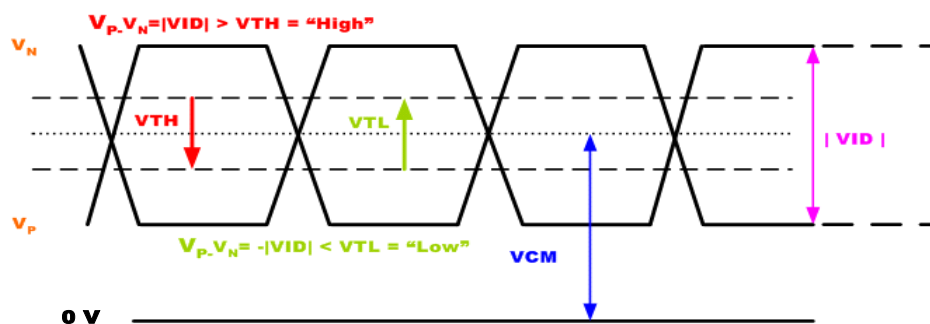
Input signals shall be low or Hi-Z state when VDD is off.

Symbol	Item	Min.	Typ.	Max.	Unit	Remark
VTH	Differential Input High Threshold	-	-	100	[mV]	VCM=1.2V
VTL	Differential Input Low Threshold	-100	-	-	[mV]	VCM=1.2V
VID	Input Differential Voltage	100	400	600	[mV]	
VICM	Differential Input Common Mode Voltage	1.1		1.45	[V]	VTH/VTL=±100mV

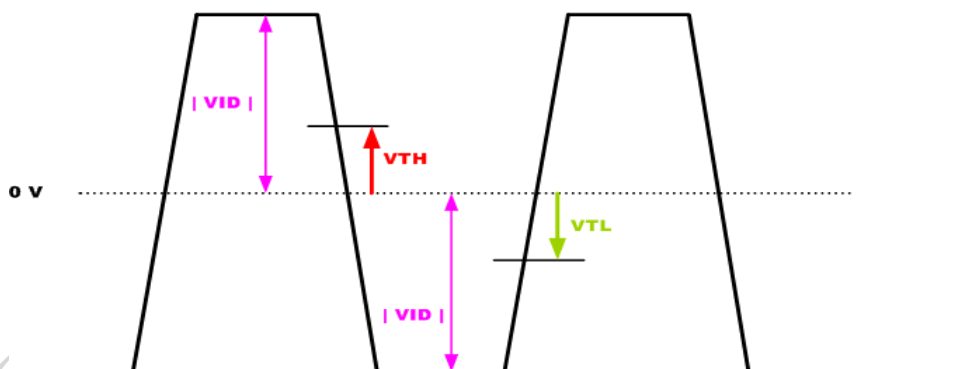
Note: LVDS Signal Waveform.

VP-VN = |VID| > VTH = "High"
VP-VN = -|VID| < VTL = "Low"

Single-end Signal



Differential Signal



5.2 Backlight Unit Driving

5.2.1 Parameter guideline for LED

Following characteristics are measured under stable condition using a LED driving board at 25°C (Room Temperature).

Symbol	Parameter	Min	Typ	Max	Units	Remark
Vcc	Input Voltage	10.8	12	13.2	Volt	
Ivcc	Input Current	-	0.58	0.65	A	100% Dimming
PLED	Power Consumption	-	6.98	7.80	Watt	100% Dimming
Irush	Inrush Current			2	A	
VLED on/off	On Control Voltage	2.5	3.3	5.5	Volt	
	Off Control Voltage			0.7	Volt	
FPWM	PWM Dimming Frequency	200		20k	Hz	
	High Voltage	2.5	3.3	5.5	Volt	
	Low Voltage			0.7	Volt	
	Dimming Duty Cycle	5	-	100	%	
I _F	LED Forward Current		60		mA	Ta = 25°C
LED Life		50,000			Hrs	Ta = 25°C

Note 1: Ta means ambient temperature of TFT-LCD module.

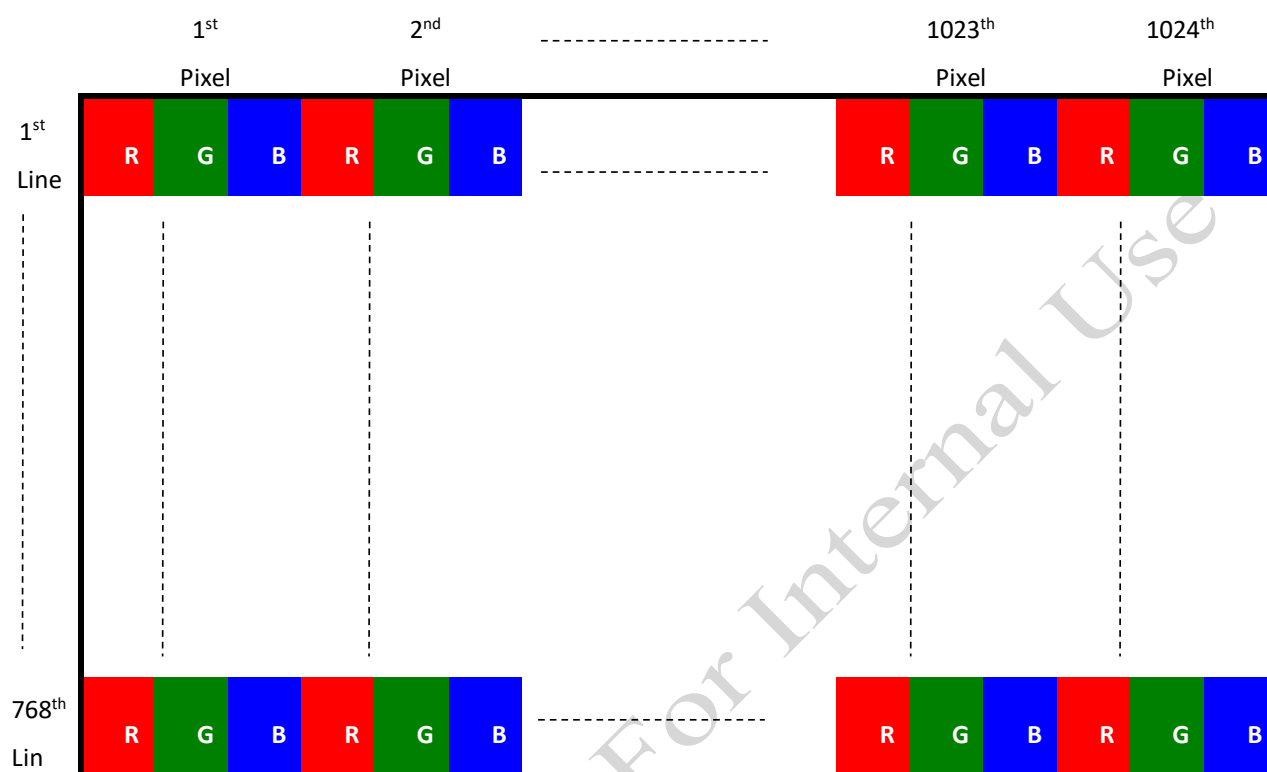
Note 2: If G150XTN06.8 module is driven at high ambient temperature & humidity condition. The operating life will be reduced.

Note 3: LED life means brightness goes down to 50% initial brightness. Min. operating life time is estimated data.

6. Signal Characteristic

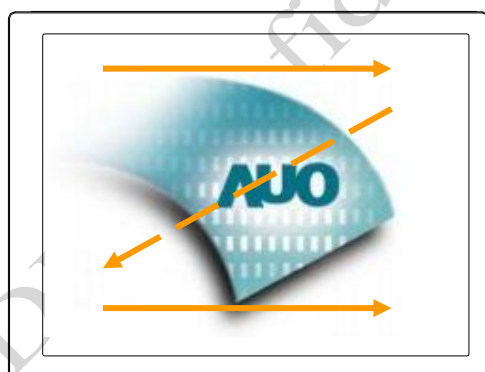
6.1 Pixel Format Image

Following figure shows the relationship between input signal and LCD pixel format.



6.2 Scanning Direction

The following figures show the image seen from the front view. The arrow indicates the direction of scan.



6.3 TFT-LCD Interface Signal Description

LVDS is a differential signal technology for LCD interface and high speed data transfer device.

Pin No.	Symbol	Description
1	VDD	Power Supply, 3.3V (typical)
2	VDD	Power Supply, 3.3V (typical)
3	NC	No Connect
4	NC	No Connect
5	Rin1-	- LVDS differential data input
6	Rin1+	+ LVDS differential data input
7	VSS	Ground
8	Rin2-	- LVDS differential data input
9	Rin2+	+ LVDS differential data input
10	VSS	Ground
11	Rin3-	- LVDS differential data input
12	Rin3+	+ LVDS differential data input
13	VSS	Ground
14	ClkIN-	- LVDS differential clock input
15	ClkIN+	+ LVDS differential clock input
16	VSS	Ground
17	Rin4-	- LVDS differential data input
18	Rin4+	+ LVDS differential data input
19	VSS	Ground
20	SEL LVDS	H or NC: 6bit/L: 8bit

6.4 Interface Timing

6.4.1 Timing Characteristics

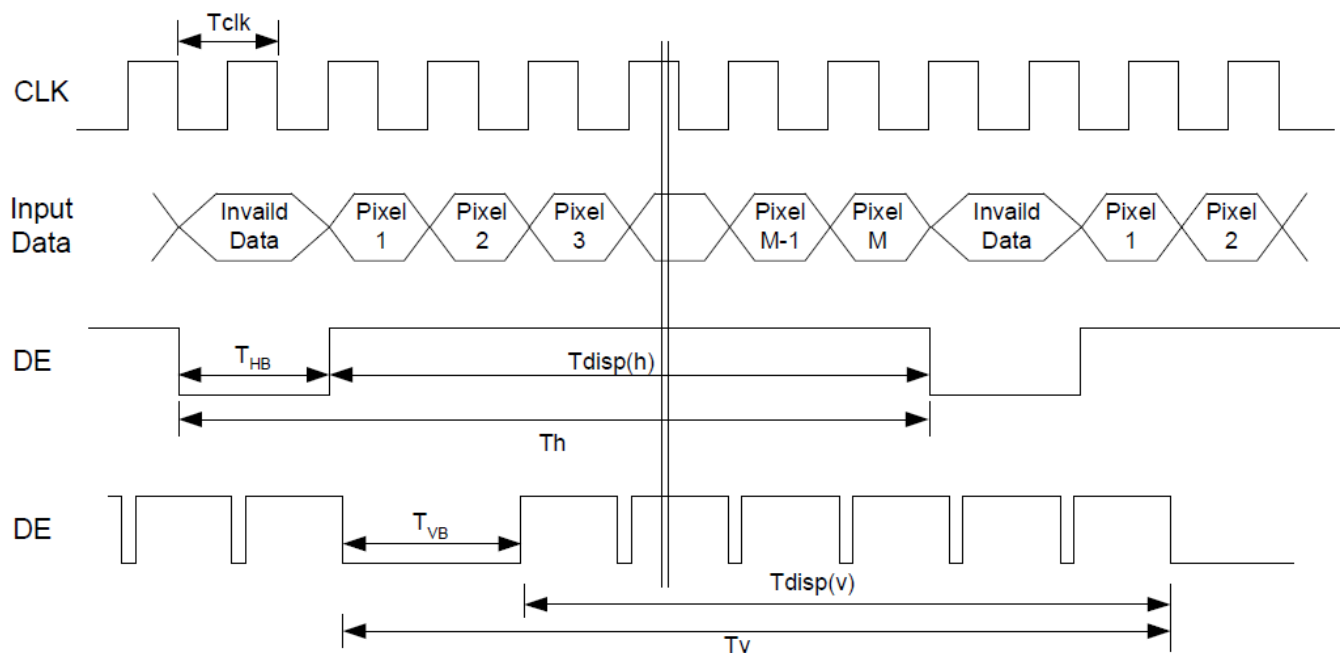
Signal	Parameter		Symbol	Min.	Typ.	Max.	Unit
Clock Timing	Clock frequency		1/ T _{Clock}	50	65	80.3	MHz
Vsync Timing	Vertical Section	Period	T _V	783	806	824	Vsync Timing
		Active	T _{VD}	768	768	768	
		Blanking	T _{VB}	15	38	56	
Hsync Timing	Horizontal Section	Period	T _H	1224	1344	1624	Hsync Timing
		Active	T _{HD}	1024	1024	1024	
		Blanking	T _{HB}	200	320	600	
Frame Rate			F		60		Hz

Note: The maximum Frame Rate $< 81\text{MHz} / [(V_Period) * (H_Period)]$

Note: DE mode only.

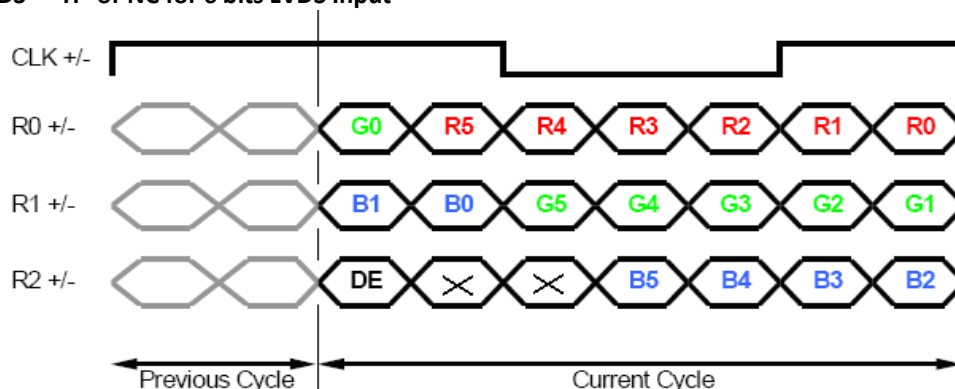
Note : Typical value refer to VESA STANDARD

Input Timing Diagram

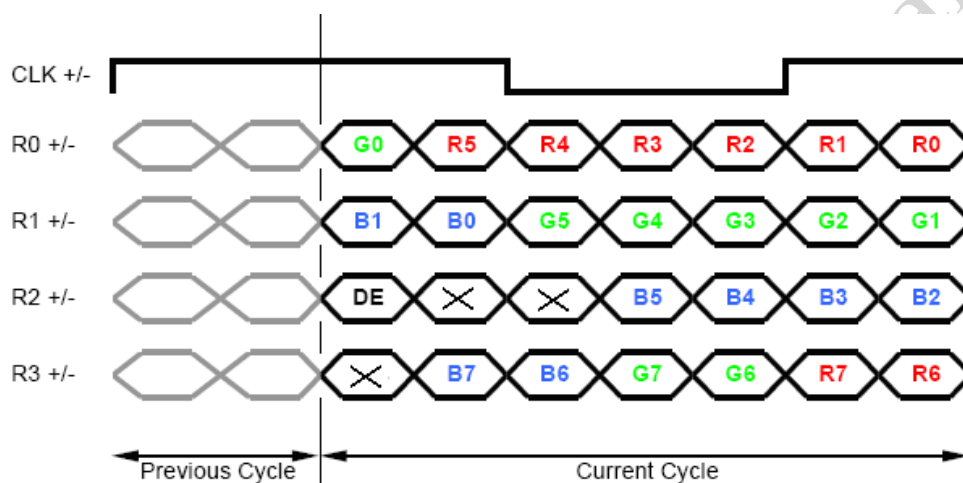


6.4.2 The Input Data Format

SEL LVDS = "H" or NC for 6 bits LVDS Input



SEL LVDS = "L" for 8 bits LVDS Input

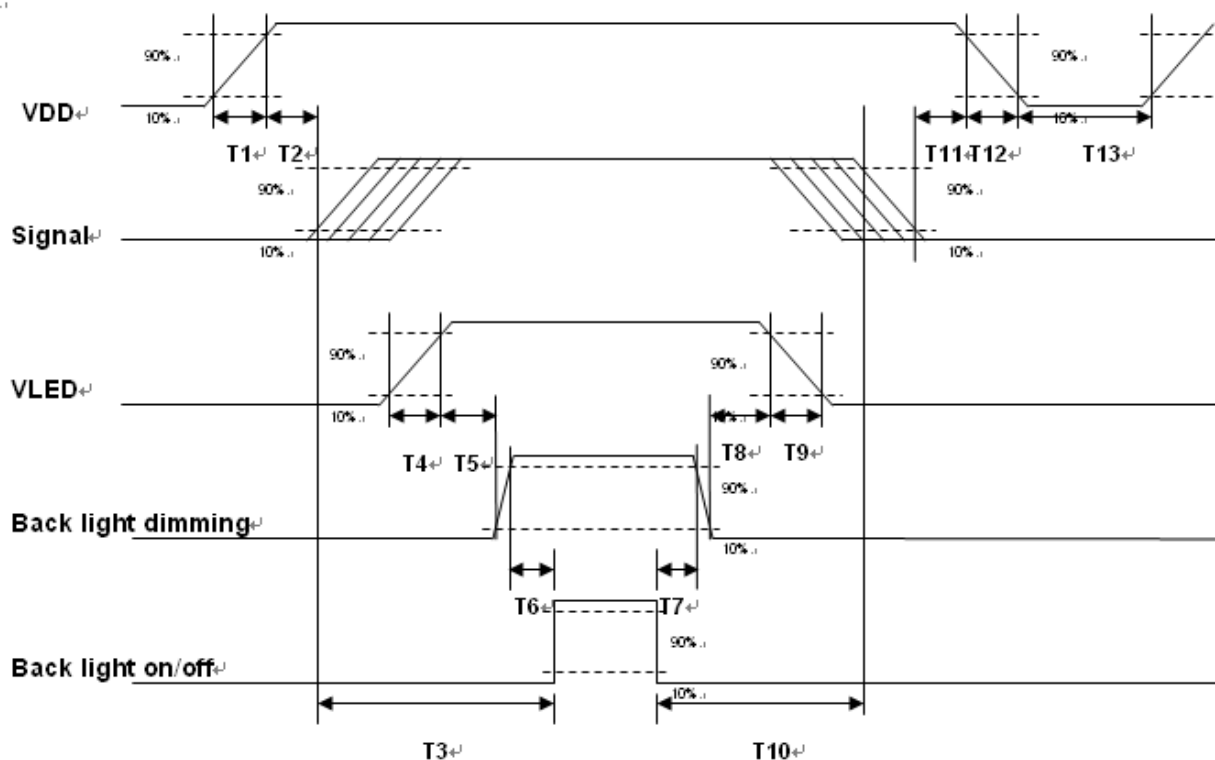


Note1: Please follow PSWG.

Note2: R/G/B data 7:MSB, R/G/B data 0:LSB

6.5 Power ON/OFF Sequence

VDD power and LED on/off sequence is as below. Interface signals are also shown in the chart. Signals from any system shall be Hi-Z state or low level when VDD is off.



Power ON/OFF sequence timing

Parameter	Value			Units
	Min.	Typ.	Max.	
T1	0.5	-	10	[ms]
T2	30	40	50	[ms]
T3	200	-	-	[ms]
T4	0.5	-	10	[ms]
T5	10	-	-	[ms]
T6	10	-	-	[ms]
T7	0	-	-	[ms]
T8	10	-	-	[ms]
T9	-	-	10	[ms]
T10	110	-	-	[ms]
T11	0	16	50	[ms]
T12	-	-	10	[ms]
T13	1000	-	-	[ms]

The above on/off sequence should be applied to avoid abnormal function in the display. Please make sure to turn off the power when you plug the cable into the input connector or pull the cable out of the connector.

7. Connector & Pin Assignment

Physical interface is described as for the connector on module. These connectors are capable of accommodating the following signals and will be following components.

7.1 TFT- LCD Signal (CN1): LCD Connector

Connector Name / Designation	Signal Connector
Manufacturer	STM or compatible
Connector Model Number	MSB240420HD or compatible
Mating Model Number	P240420 or compatible

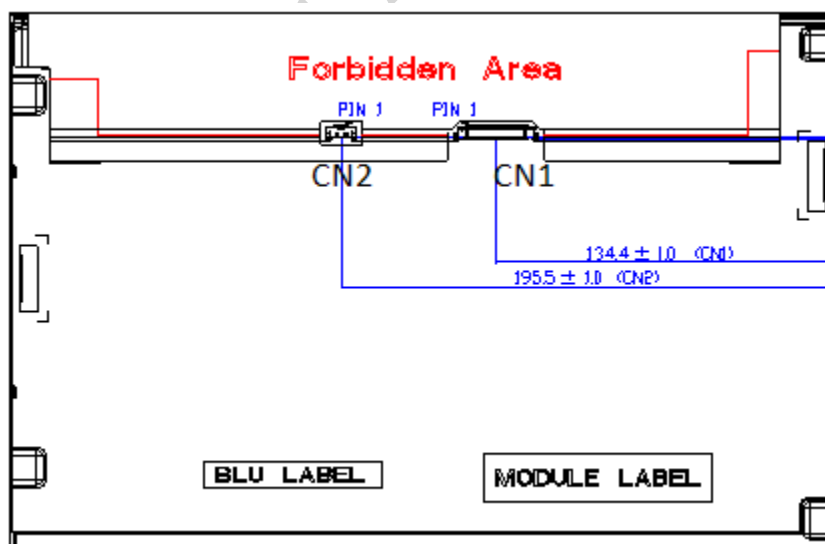
Pin#	Signal Name	Pin#	Signal Name
1	VDD	2	VDD
3	NC	4	NC
5	Rin1-	6	Rin1+
7	VSS	8	Rin2-
9	Rin2+	10	VSS
11	Rin3-	12	Rin3+
13	VSS	14	ClkIN-
15	ClkIN+	16	VSS
17	Rin4-	18	Rin4+
19	VSS	20	SEL68

7.2 LED Backlight Unit (CN2): Backlight Connector

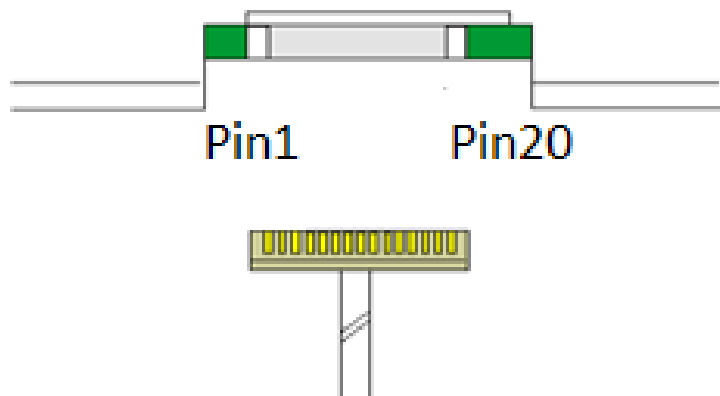
Connector Name / Designation	LED Connector
Manufacturer	STM or compatible
Connector Model Number	MSB24038P5D or compatible
Mating Model Number	P24038P5 or compatible

Pin#	Symbol	Signal Name
1	Vcc	12V
2	GND	GND
3	Enable	VLED On/Off
4	Dimming	PWM Dimming
5	NC	NC

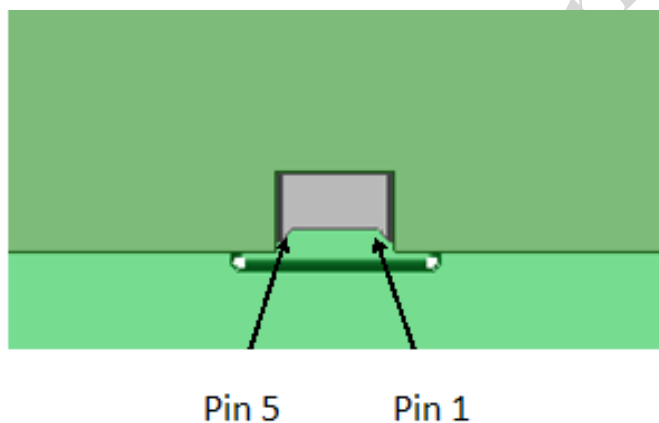
7.3 PIN 1 definition of CN1 and CN2



LVDS(CN1):



BLU Power in(CN2):



8. Reliability Test Criteria

Items	Required Condition	Note
Temperature Humidity Bias	50°C /80%,300 hours	
High Temperature Operation	85°C ,300 hours	
Low Temperature Operation	-30°C ,300 hours	
Hot Storage	85°C ,300 hours	
Cold Storage	-30°C ,300 hours	
Thermal Shock Test	-20°C /30 min ,60°C /30 min ,100cycles	
Shock Test (Non-Operating)	50G,20ms,Half-sine wave,(±X, ±Y, ±Z)	
Vibration Test (Non-Operating)	1.5G, (10~200Hz, P-P), 30 mins/axis (X, Y, Z)	
On/off test	On/10 sec, Off/10 sec, 30,000 cycles	
ESD	Contact Discharge: ± 8KV, 150pF(330Ω) 1sec, 8 points, 25 times/ point Air Discharge: ± 15KV, 150pF(330Ω) 1sec, 8 points, 25 times/ point	Note 1

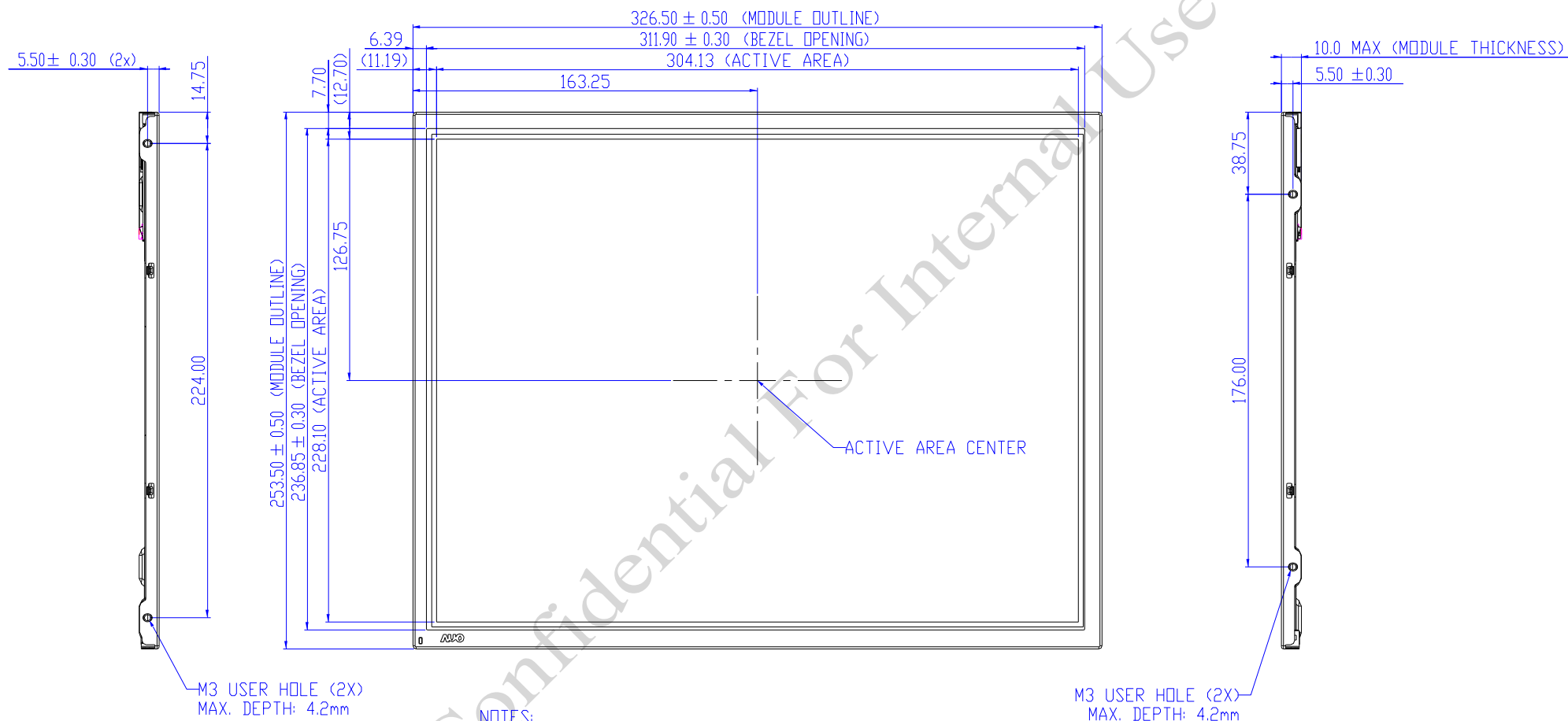
Note1: According to EN61000-4-2, ESD class B: Some performance degradation allowed. No data lost

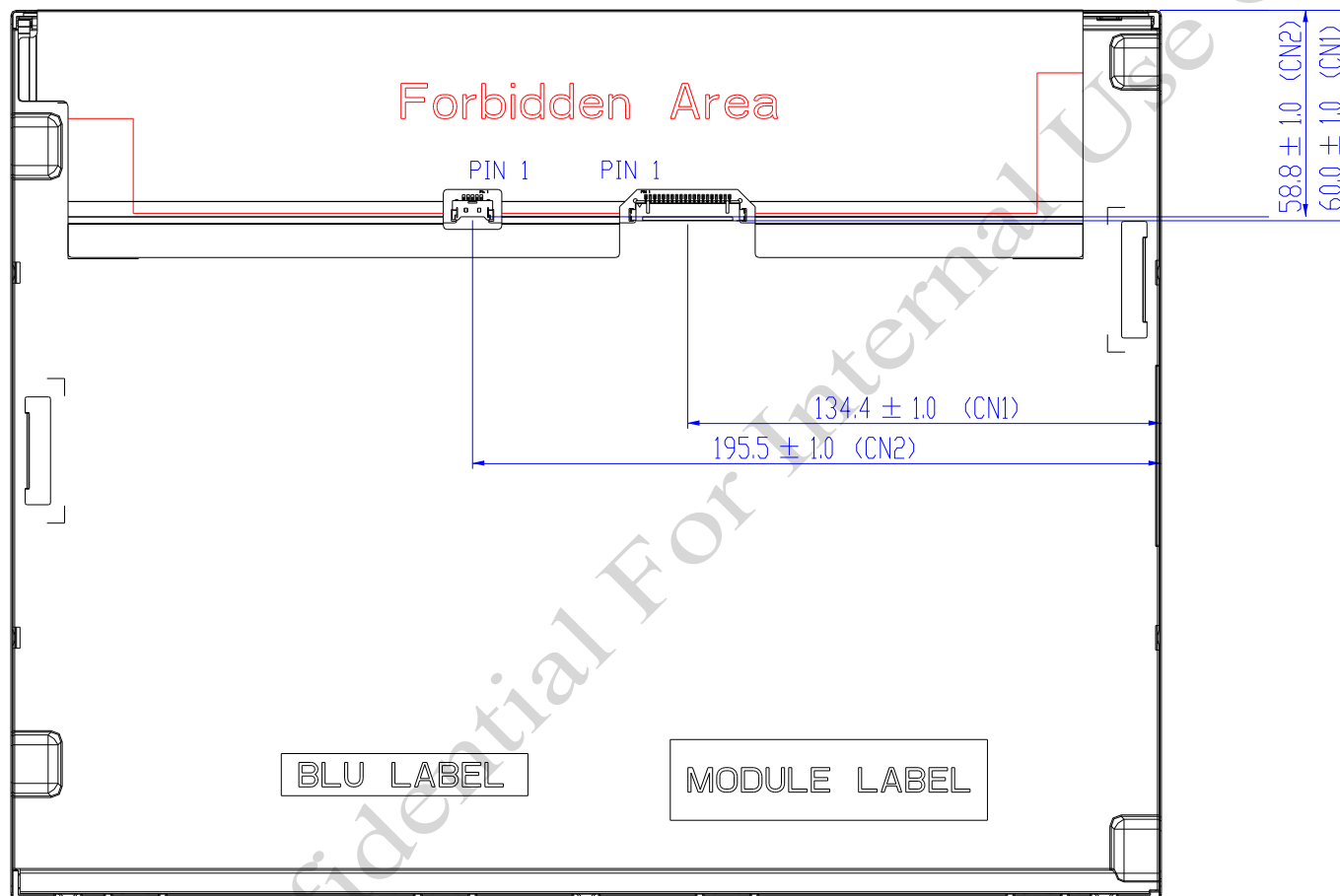
. Self-recoverable. No hardware failures.

Note 2:

- Water condensation is not allowed for each test items.
- Each test is done by new TFT-LCD module. Don't use the same TFT-LCD module repeatedly for reliability test.
- The reliability test is performed only to examine the TFT-LCD module capability.
- To inspect TFT-LCD module after reliability test, please store it at room temperature and room humidity for 24 hours at least in advance.
- No function failure occurs. Mura shall be ignored after high temperature reliability test

9. Mechanical Characteristics





10. Label and Packaging

10.1 Shipping Label (on the rear side of TFT-LCD display)



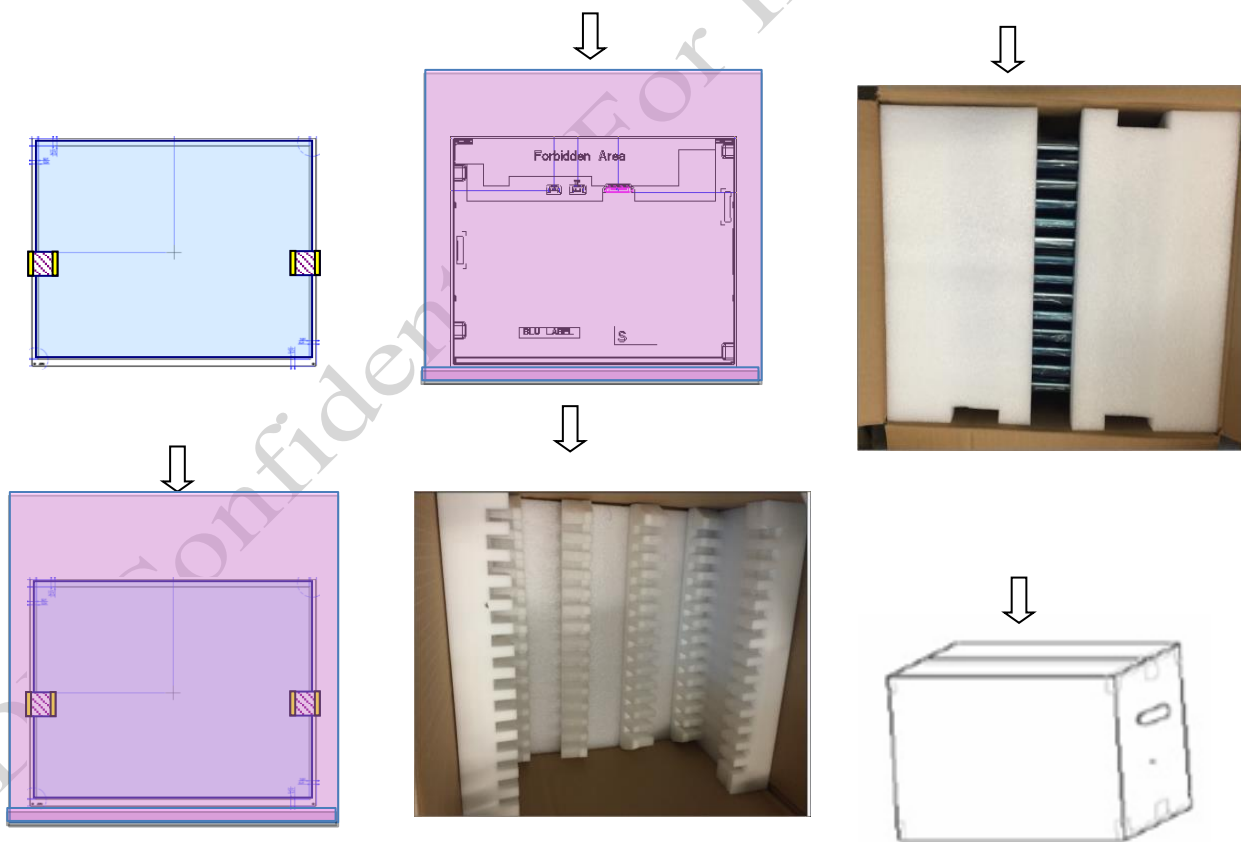
10.2 Carton Package

Max capacity : 14 TFT-LCD module per carton

Max weight: 16.5 kg per carton

Outside dimension of carton: 430mm(L)* 370mm(W)*350mm(H)

Pallet size : 1140 mm * 890 mm * 132mm



14pcs/Carton

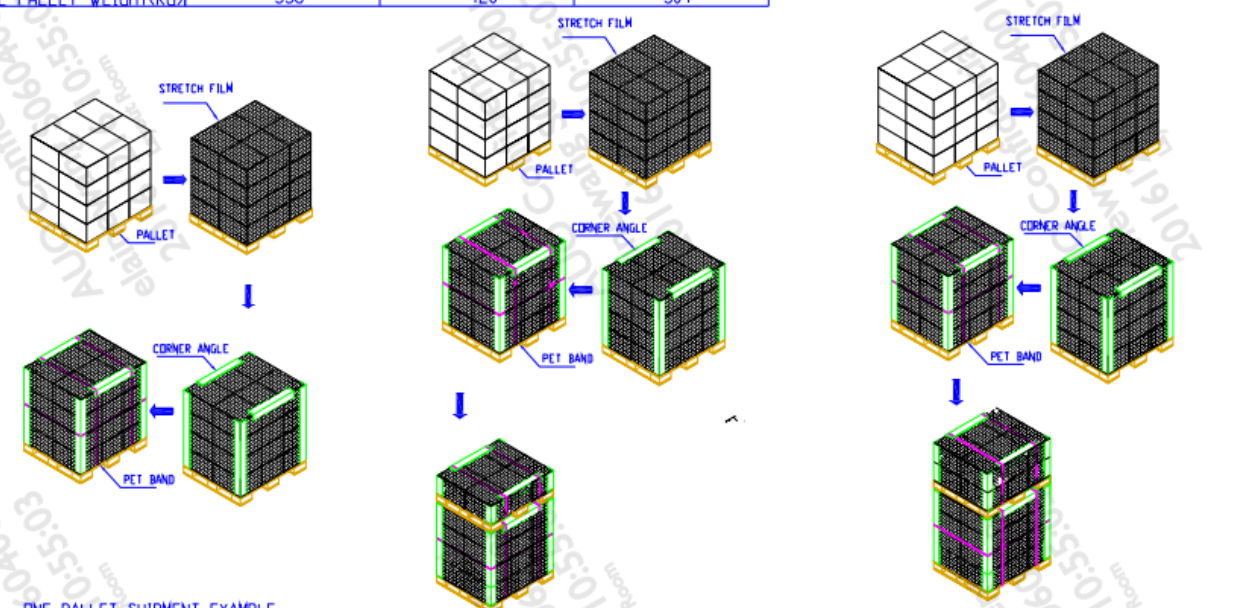
Box stacked

Module by air_Max: (2 *3) *4 layers , one pallet put 24 boxes , total 336pcs module

Module by sea_Max: (2 *3) *4 layers + (2 *3) *1 layers , two pallet put 30 boxes , total 420pcs module

Module by sea_HQ_Max: (2 *3) *4 layers+(2 *3) *2 layers, two pallet put 36 boxes, total 504pcs module

	SHIPPING BY AIR	SHIPPING BY SEA	SHIPPING BY SEA_HQ
MODULE/CARTON	14	14	14
CARTON/LAYER	6	6	6
LAYER/PALLET	4	4+1	4+2
MODULE/PALLET	336	420	504
CARTON SIZE(MM)	430(L)*370(W)*350(H)	430(L)*370(W)*350(H)	430(L)*370(W)*350(H)
PALLET SIZE(MM)	1140(L)*890(W)*132(H)	1140(L)*890(W)*132(H)	1140(L)*890(W)*132(H)
TOTAL PALLET HEIGHT(MM)	1535	2020	2370
TOTAL PALLET WEIGHT(KG)	336	420	504



ONE PALLET SHIPMENT EXAMPLE SHIPPING BY AIR USED

TWO PALLETS SHIPMENT EXAMPLE SHIPPING BY SEA USED

TWO PALLETS SHIPMENT EXAMPLE SHIPPING BY SEA_HQ USED

11 Safety

11.1 Keen Edge Requirements

There will be no keen edges or corners on the display assembly that could cause injury.

11.2 Materials

11.2.1 Toxicity

There will be no carcinogenic materials used anywhere in the TFT-LCD module. If toxic materials are used, they will be reviewed and approved by the responsible AUO toxicologist.

11.2.2 Flammability

The printed circuit board will be made from material rated 94-V1 or better. The actual UL flammability rating will be printed on the printed circuit board.

11.3 Capacitors

If any polarized capacitors are used in the TFT-LCD module, provisions will be made to keep them from being inserted backwards.

11.4 International Safety Standard Compliance

The TFT-LCD Module will satisfy all requirements for compliance to IEC/UL 62368-1.